Indian Institute of Technology Guwahati
Department of Energy Engineering

B.TECH PROJECT · FINAL PRESENTATION

Optimal Cost Sizing of a PV–Battery Microgrid for IIT Guwahati

Lifecycle economics, reliability and robustness analysis

PRESENTED BY

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Roll No. [add before presenting]

UNDER THE GUIDANCE OF

Kuldeep Kumar

[Presentation date]

ROADMAP

Outline of the talk

01 Slides 3–6

Introduction

Microgrids, IITG demand, solar resource, objectives

02 Slides 7–11

Literature and gap

Landscape, reference study, comparison, contribution

03 Slides 12–18

Modelling and optimisation

Architecture, data, component and cost models, PSO

04 Slides 19–22

Optimal design

Sizing, design space, dispatch, battery ageing

05 Slides 23–26

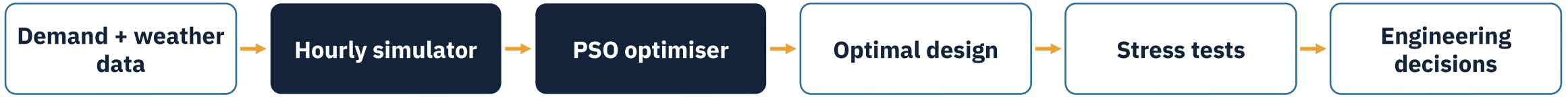
Robustness

Weather years, sensitivity, reliability target, Monte Carlo

06 Slides 27–34

Scenarios and conclusions

Load growth, grid access, supercapacitor, CERC check

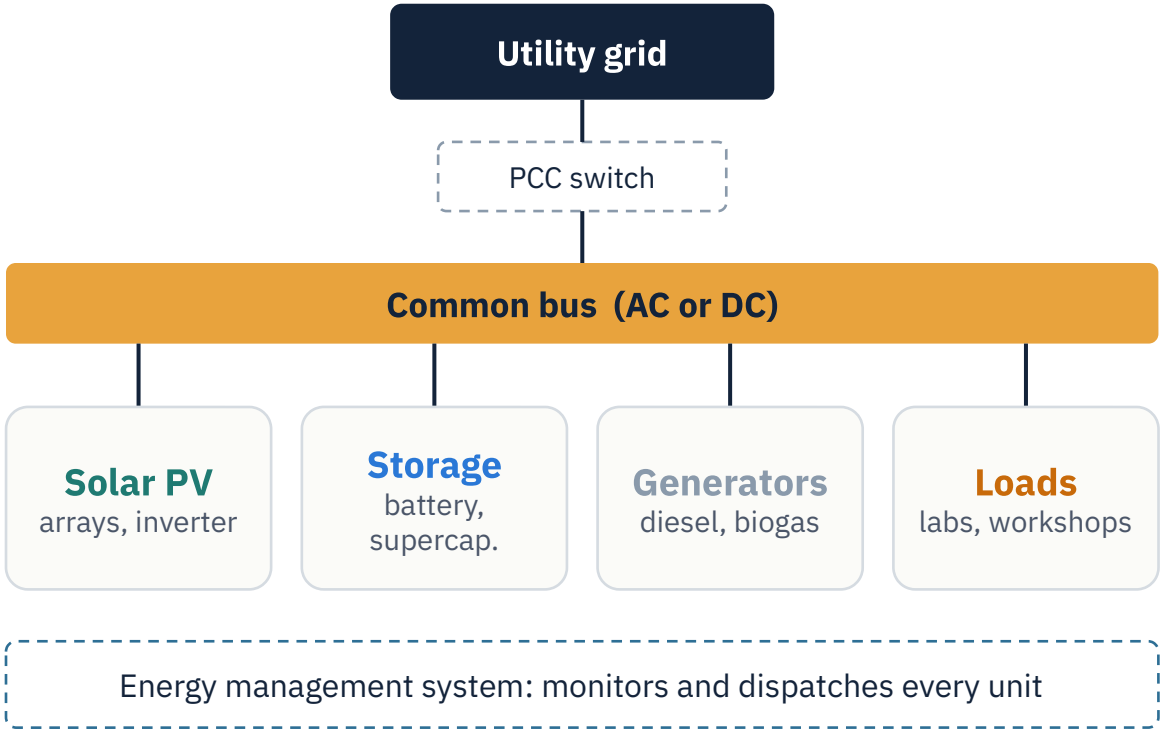


The logic of the project: every result in Parts IV–V comes from the same verified simulator.

What is a microgrid?

A microgrid is a **local group of generators, storage and loads** controlled as one unit — connected to the main grid or running on its own (islanded).

- Generation sits close to the load
- Storage covers the gap between supply and demand
- A controller dispatches every unit, hour by hour
- Islanded operation: every kWh must come from local sources



AC-coupled · this project

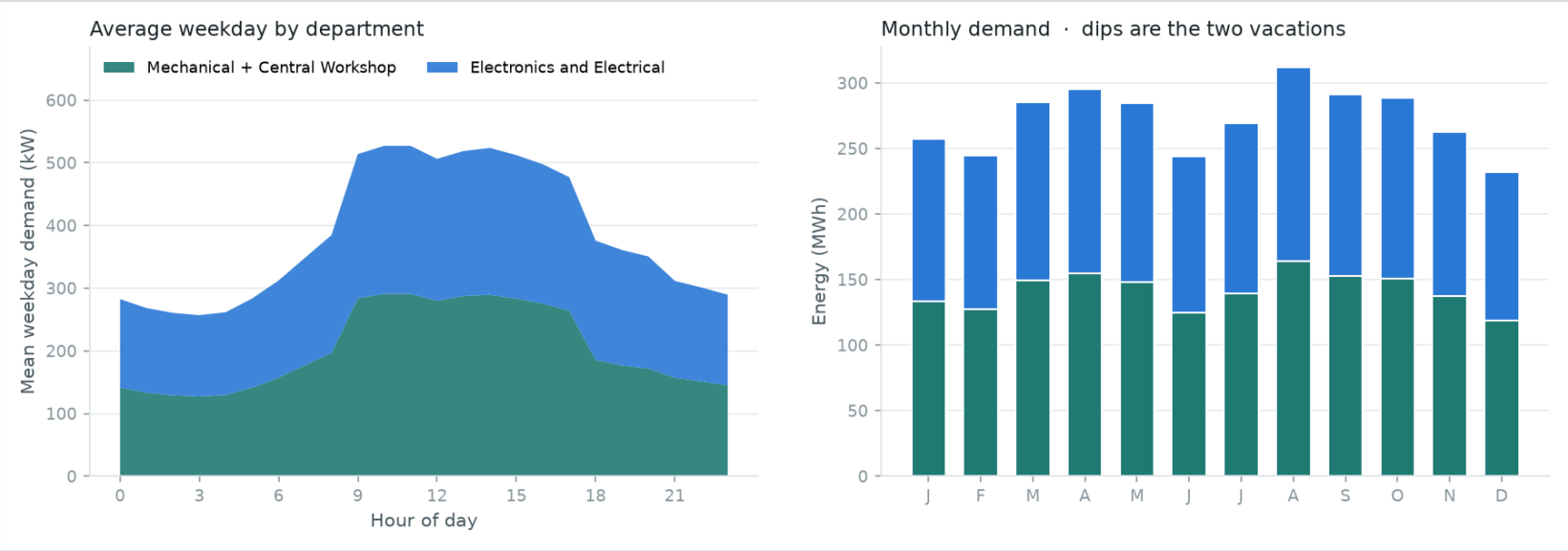
PV inverter and battery converter each feed an AC bus. Same topology as Kumar et al. (2018).

DC-coupled

PV and batteries share a DC bus: fewer conversion stages, natural for storage and supercapacitors.

Lasseter (2002); Hatziargyriou et al. (2007); Dragičević et al. (2016).

IIT Guwahati departmental electricity demand



Average weekday by department, and monthly energy (dips = vacations). Source: prepare_data.py.

3,267 MWh
annual demand

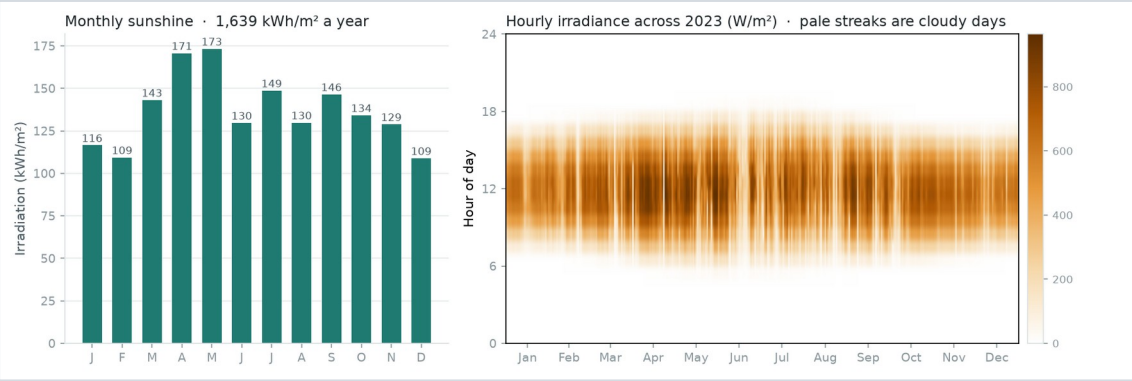
644 kW
peak demand

0.58
load factor (mean / peak)

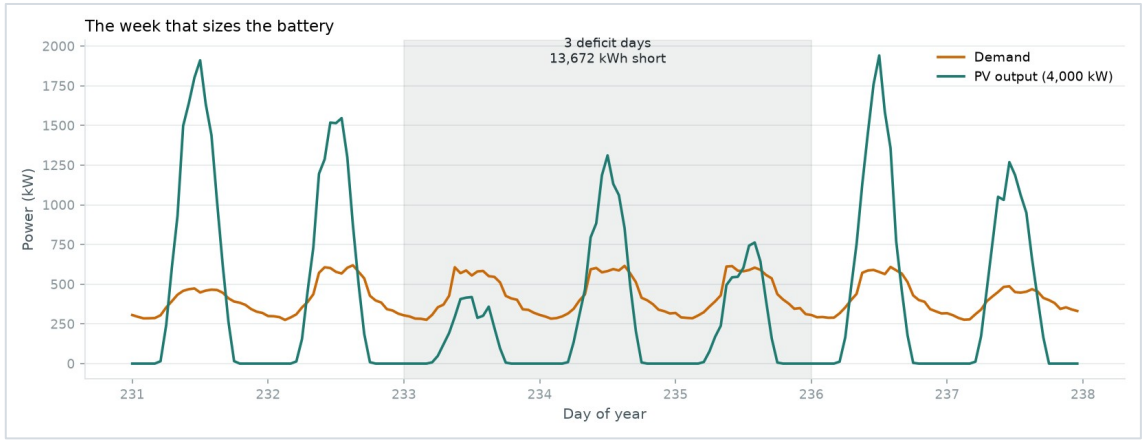
How the hourly load was built · assumption, not metered data

- Equipment inventory per department → rated kW × duty × operating hours → weekday and two-vacation calendar → hourly profile, calibrated against campus statistics (Kumar et al. 2018).
- Mechanical + Workshop 1,702 MWh (52 %) · Electronics + Electrical 1,565 MWh (48 %). Power ratings are engineering estimates.

Solar resource and why storage is needed



NASA POWER, 26.16 °N 91.77 °E, 2023.



Worst week: PV (teal) against demand (amber).

1,639 kWh/m²
global horizontal irradiation, 2023

12 / 12
months in which PV energy exceeds demand

3 days
dull spell, ≈13.7 MWh short: this sizes the battery

Takeaway Storage is needed for nights and multi-day cloudy spells, not for seasons.

Research motivation and objectives

Cost

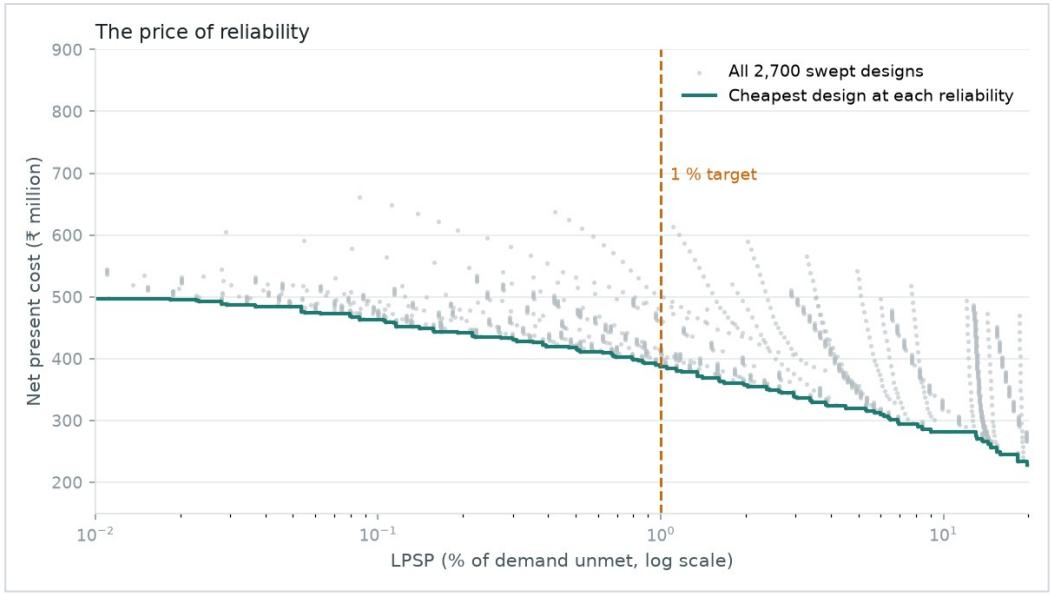
PV is cheap per kWh but only produces in daylight; batteries and replacements dominate later costs.

Reliability

Each step towards zero unmet demand needs more storage and PV, at a rising price.

Lifetime

Batteries wear out within the 15-year horizon, so ageing has to be modelled, not assumed.

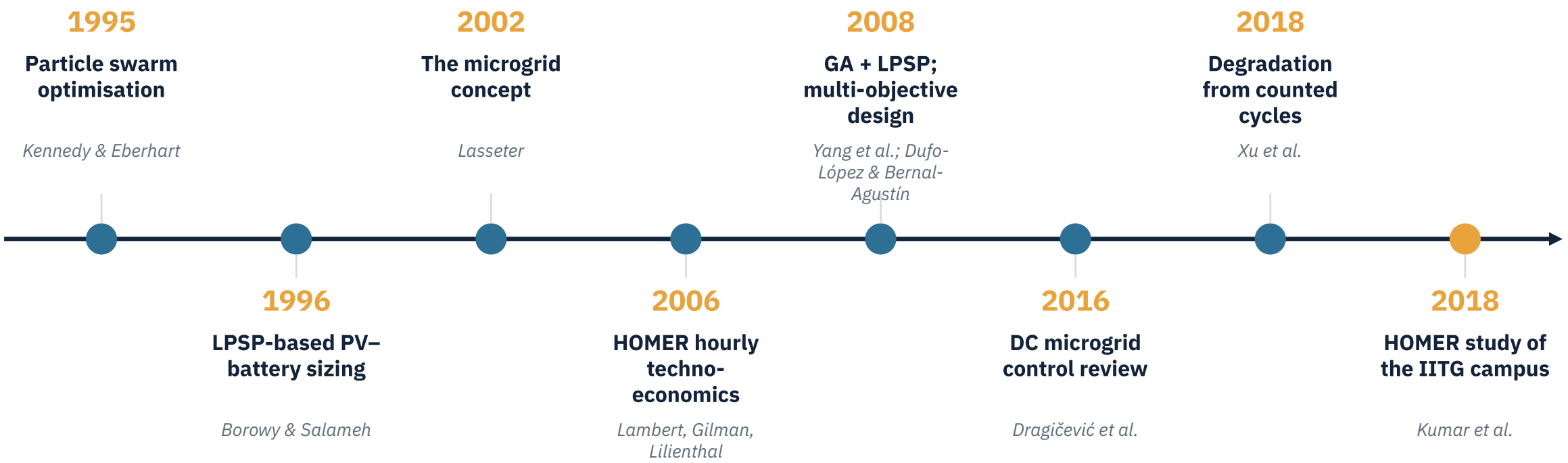


Cheapest design at each reliability level (grid sweep).

Objective

Find the least-cost PV–battery configuration that supplies both departments with **unmet demand ≤ 1 %**, over a **15-year lifecycle** — then test that answer against weather, demand growth, grid access, a supercapacitor and uncertain inputs.

Research landscape



Sizing for reliability | **Lifecycle economics** | **Battery ageing** | **Weather and uncertainty** | **Metaheuristic search**

Themes this project combines. Full references on the last slide.

Reference study: Kumar et al. (2018), IEEE Access

What they did

- HOMER Pro, multi-year module
- Whole IITG campus: peak 3,075 kW, ≈60,379 kWh/day
- 12 scenarios: PV, biogas, diesel, lead-acid, grid
- 10 % yearly load growth, 15-year project

What this project takes

- Component costs (Table 2) and economics
- Discover 12VRE-3000TF-L lead-acid battery
- PV model (eq. 3) and AC-coupled topology
- Benchmarks for comparison

What this project changes

- Two departments, built bottom-up
- Own Python simulator with 35 checks
- Rainflow ageing; PSO + grid search
- 12 weather years; Monte Carlo

₹5,835 M
best off-grid NPC, whole campus (scenario 8)

₹11.96–18.47
off-grid LCOE range, per kWh

₹0.18–1.39
grid-connected LCOE range, per kWh

All values from Kumar et al. (2018), checked against the paper's text. Whole-campus figures, so they are benchmarks for unit cost, not for absolute NPC.

Critical literature comparison

Study	System	Method	Objective / reliability	What it contributes
Borowy & Salameh (1996)	Stand-alone wind–PV–battery	Iterative sizing of PV and battery	Minimum cost at a target LPSP	30 years of hourly data; house load
Yang et al. (2008)	Stand-alone PV–wind–battery	Genetic algorithm	Minimum annualised cost at a target LPSP	Telecom relay station case
Dufo-López & Bernal-Agustín (2008)	PV–wind–diesel–H_2–battery	Multi-objective evolutionary + GA	NPC, CO_2 and unmet load together	Pareto set of designs
Xu et al. (2018)	Li-ion battery storage	Semi-empirical model with cycle counting	Cell life loss from operating profile	Degradation from irregular cycles
Kumar et al. (2018)	IITG campus: PV, biogas, diesel, battery, grid	HOMER Pro multi-year	NPC and LCOE, 12 scenarios	Site data, costs, benchmark
This work	Two IITG departments: PV–battery (+ SC option)	Own simulator; grid search + PSO	Min. 15-year NPC, LPSP ≤ 1 %	Rainflow ageing, 12 weather years, Monte Carlo

LPSP: loss of power supply probability. SC: supercapacitor. Details confirmed from publisher or index records (DOIs in References).

Research gap and contribution

Gaps in the reference approach

Battery life as constant throughput

HOMER's default model ignores depth of discharge.

One weather year

A design can look reliable only because the year was sunny.

Campus aggregate load

Department-level demand and its uncertainty are not studied.

Unchecked inputs

Reference costs are reused without going back to the source.

What this project adds

Cycle-counted ageing

Rainflow + Miner's rule on the datasheet curve, plus temperature float life.

Weather robustness

Same design on 12 NASA years; a design optimised for all of them.

Bottom-up demand

Two departments from equipment inventories; tested ±20 %.

Verified, open model

35 automated checks; PSO cross-checked by grid search.

Scenario studies

Staged growth, grid access, supercapacitor economics.

Source audit

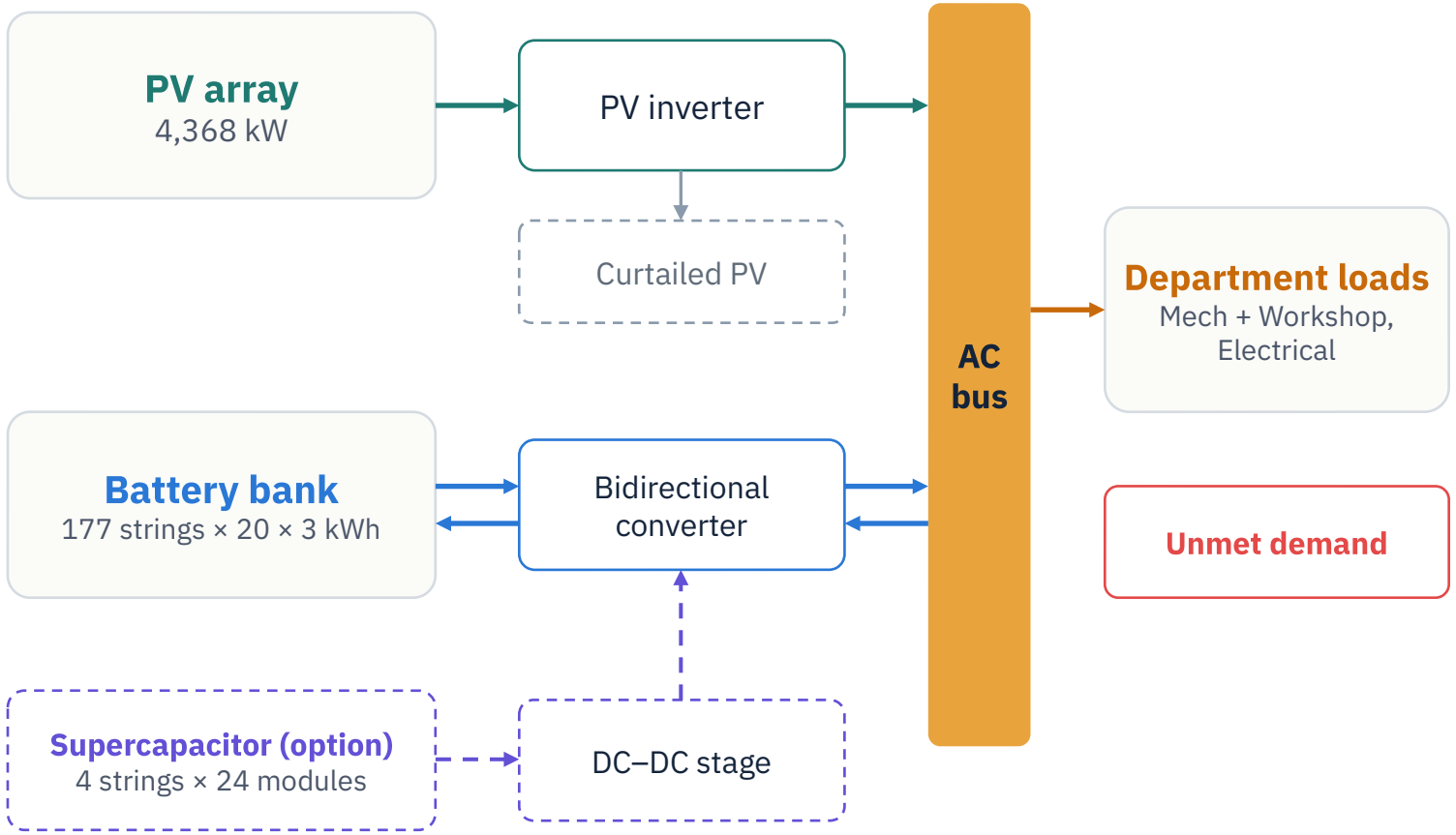
CERC order checked: paper's PV cost and discount rate corrected.

METHODOLOGY

Research questions, methods and answers

#	Research question	Method (script)	Short answer	Slide
1	Does PSO reproduce the grid-search optimum?	Grid sweep + PSO, 5 seeds (pso.py)	Yes: 5/5 runs within 0.41 %, below the grid best	18–19
2	How does battery ageing affect lifecycle cost?	Rainflow + Miner (simulator.py)	Life 7.7 y; replacements ₹54.6 M	22
3	Is the design reliable in other weather years?	12 NASA years (performance.py)	No: misses 1 % in 8 of 12 years	23
4	Does staged expansion save cost under growth?	15-year growth model (multiyear.py)	Yes: 33 % cheaper at 10 %/yr	27
5	How does grid access change the design?	Grid scenarios (grid_connected.py)	Grid + 2,400 kW PV, no battery: ₹142 M	28
6	Is a supercapacitor economically justified?	1-s load + filter (sc_study.py)	At best break-even ($\tau \approx 1$–2 s)	29–30
7	Which inputs drive cost and reliability?	Tornado, spider (sensitivity.py)	Demand, then PV and battery prices	24–25
8	How uncertain are the results?	Monte Carlo, 1,000 runs	NPC ₹343 M–₹436 M; target missed 60 %	26
9	Do corrected CERC inputs change the design?	Re-run with CERC values	NPC +2.3 %, design unchanged	31

Proposed system architecture



Studied separately with 1-second load data (slides 29–30); not part of the hourly balance.

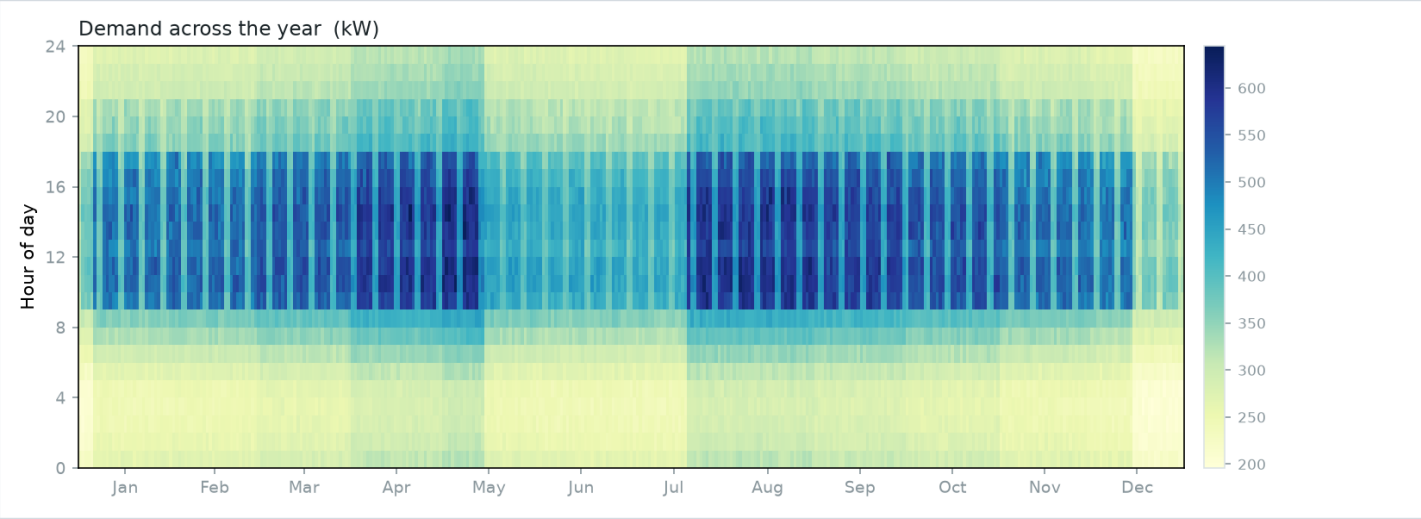
Hourly dispatch rules

1. PV serves the load directly.
2. Surplus charges the battery, limited by charge current, converter rating and room in the battery.
3. Whatever cannot be stored is curtailed.
4. A deficit is drawn from the battery, down to 20 % state of charge.
5. Demand the battery cannot cover is unmet: it counts towards LPSP.

Demand and weather data pipeline

Equipment inventory per department → Rated kW × duty × operating hours → Weekday and vacation calendar → Hourly department profiles → **data/inputs.csv** 8,760 hours: load, GHI, T

NASA POWER 2023 GHI + temperature → Gap filling and clean-up → **data/inputs.csv**



Every hour of the year: working hours, weekends and vacations show clearly.

Worst-case transient → transient_spec.csv
4 machines together: 93 kW for 5 s (465 kJ)

NASA POWER 2013–2024
weather years for robustness and Monte Carlo

Assumptions to state

- Ratings are estimates; load is not metered
- Same 2023 load shape in every weather year

PV, battery and converter models

PV array

$$P_{PV}(t) = P_r \cdot f_d \cdot G(t)/G_{STC} \cdot [1 + \alpha (T_c(t) - 25)]$$

$$T_c(t) = T_a(t) + (NOCT - 20)/800 \cdot G(t)$$

- $f_d = 0.85$ derating
- $\alpha = -0.40$ %/°C
- NOCT = 45 °C, $G_{STC} = 1{,}000$ W/m²
- Inverter inside CERC all-in cost

Battery bank

$$E(t+1) = E(t) + \eta_b \cdot P_{ch}(t)$$

$$E(t+1) = E(t) - P_{dis}(t) / \eta_b$$

$$0.2\,E_{max} \le E(t) \le E_{max}$$

- $\eta_b = \sqrt{0.85}$ each way
- 3 kWh per battery, strings of 20
- $P_{ch} \le 0.684$ kW, $P_{dis} \le 1.607$ kW per battery
- (57 A and 133.9 A at 12 V)

Bidirectional converter

$$P_{ch} = \min(\text{surplus}\cdot\eta_c, P_{conv}\cdot\eta_c, P_{ch,max}, \text{room}/\eta_b)$$

$$P_{dis} = \min(\text{deficit}/\eta_c, P_{conv}/\eta_c, P_{dis,max}, \text{avail})$$

- $\eta_c = 0.95$ each way
- Optimised rating: 998 kW
- AC side never exceeds the rating (checked)

Reliability and lifecycle cost formulation

Reliability and economics

$$LPSP = \Sigma E_{unmet}(t) / \Sigma E_{load}(t) \leq 1\ \%$$

$$NPC = C_{cap} + \Sigma C_{rep,k}/(1+r)^{t_k} + \Sigma C_{O\&M}/(1+r)^{t} - C_{salv}/(1+r)^{15}$$

$$r = (1 + i_{nom})/(1 + f) - 1 = 1.1070/1.0571 - 1 = 4.72\ \%$$

$$CRF = r(1+r)^N / ((1+r)^N - 1) = 0.0945, \quad LCOE = NPC \cdot CRF / E_{served}$$

- Salvage: last unit bought × fraction of life left at year 15 (linear, as HOMER)
- 15-year horizon; costs in constant 2016–17 rupees (Kumar et al., CERC)
- LPSP counts energy not served, not the number of shortage hours

Battery ageing

1. Rainflow-count the hourly state of charge → cycles of depth D_i
2. Cycle life N(D) from the datasheet: 5,000 at 20 %, 2,500 at 50 %, 1,500 at 80 % DoD
3. Miner's rule: damage per year = $\Sigma n_i / N(D_i)$; wear life = 1 / damage
4. Float life from service life vs temperature: 12.1 years at this site
5. Battery life = min(wear life, float life) → replacements and salvage

Mathematical optimisation formulation

Decision vector

x = [P_PV (kW), N_str (integer), P_conv (kW), N_SC (integer)]

Minimise

NPC(x) over 15 years

Subject to

LPSP(x) ≤ 0.01

N_SC ≥ N_SC,req = 4   (465 kJ transient)

x within the bounds in the table

Variable	Lower	Upper	Unit
PV capacity	1,000	8,000	kW
Battery strings	0	500	× 60 kWh
Converter	300	3,000	kW
SC strings	4	4	fixed at need

Bounds are wider than the grid sweep, so PSO is not boxed in.

Feasibility-first score

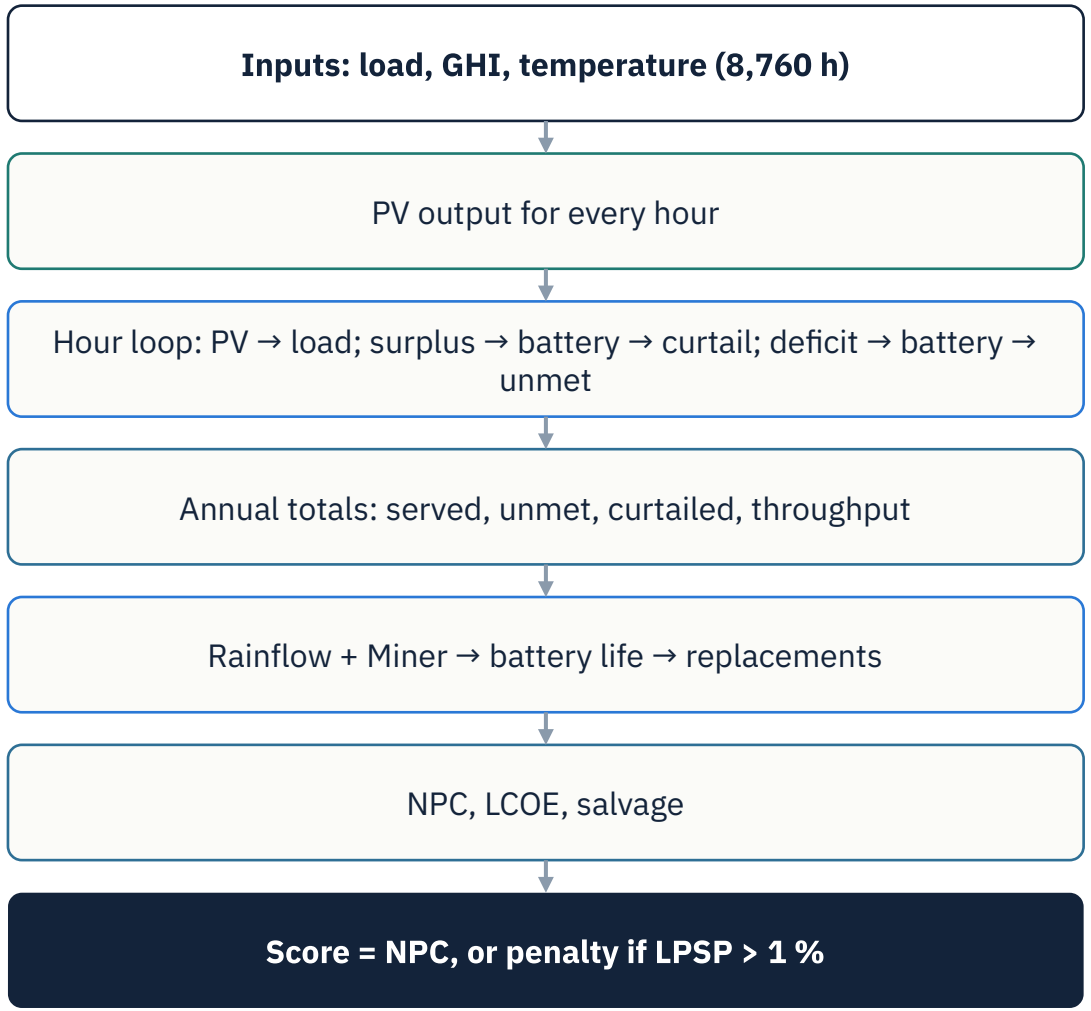
score = NPC if feasible

score = $10^{10} \cdot (1 + \text{violation})$ otherwise

violation = max(0, LPSP − 0.01) + max(0, N_SC,req − N_SC)

- Every feasible design beats every infeasible one
- Among infeasible designs, the smaller violation wins
- Avoids NPC + k·violation, where a cheap unreliable design can win

Python simulator and its verification



Verification: 35 checks, all passing

PASS

Simulator · 20

Energy balances close; SoC, converter and current limits hold; LCOE, salvage, cycle and float life match hand calculations; penalties rank correctly

PASS

Multi-year model · 7

With no growth the 15-year model equals the single-year one for three designs; yearly costs add up to NPC

PASS

Grid model · 8

Grid-only cost matches the hand calculation; energy balances close; exports never exceed the cap

Verification, not validation: the code is checked against itself and hand calculations, not against a measured microgrid.

Grid search and particle swarm optimisation

Initialise 20 particles at random within bounds

↓

Round to whole kW / strings; simulate (cached)

↓

Update personal and global best scores

↓

$v \leftarrow w \cdot v + c_1 r_1 (p_best - x) + c_2 r_2 (g_best - x)$

↓

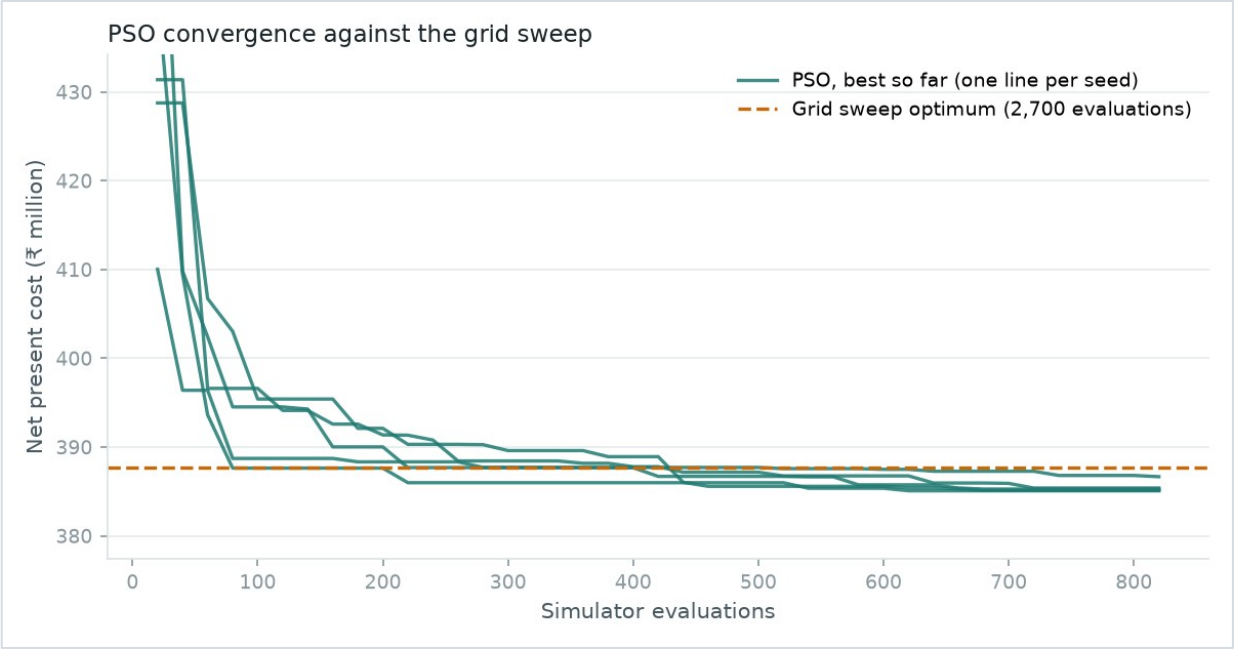
Clip $|v| \le 20\ \%$ of range; walls absorb

↓

Repeat 40 iterations → best design

loop

$w = 0.7298$, $c_1 = c_2 = 1.49618$ (Clerc & Kennedy 2002).



Best NPC so far for each of 5 seeds, against the grid-sweep optimum.

2,700
grid designs; best ₹387.7 M

5 / 5
PSO seeds within 0.41 %

≈820
simulations per PSO run

RESULTS

Optimal off-grid sizing

4,368 kW
PV array

10.6 MWh
battery (177 strings)

998 kW
converter

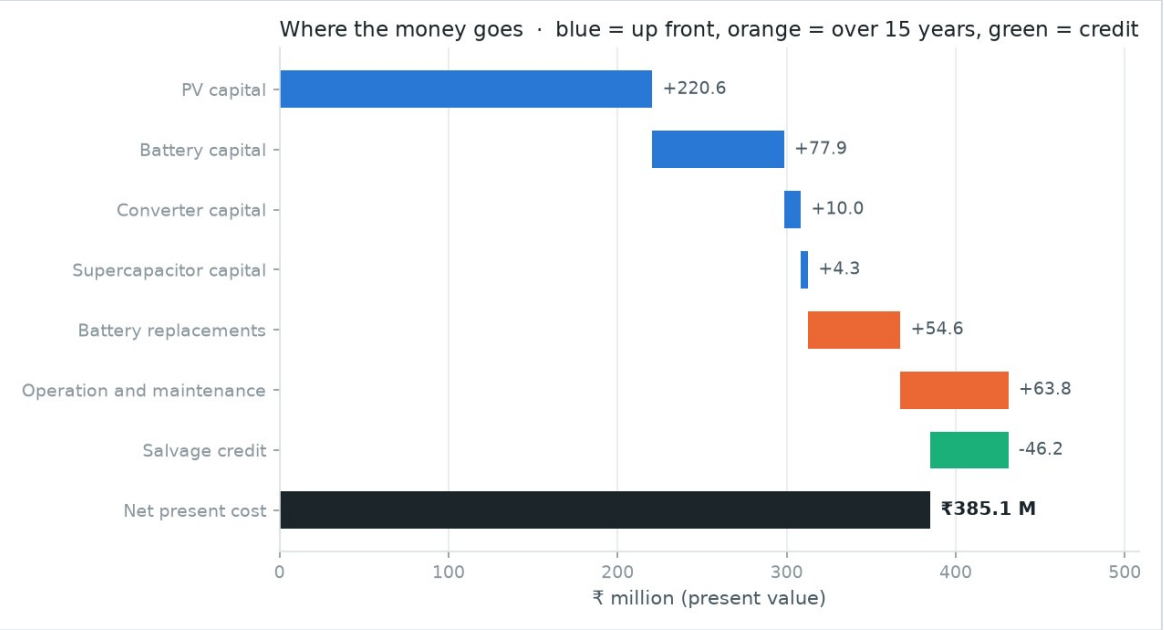
4 strings
supercapacitor (required)

₹385.1 M
15-year NPC

₹11.25/kWh
LCOE

1.00 %
LPSP (limit 1 %)

7.7 years
battery life

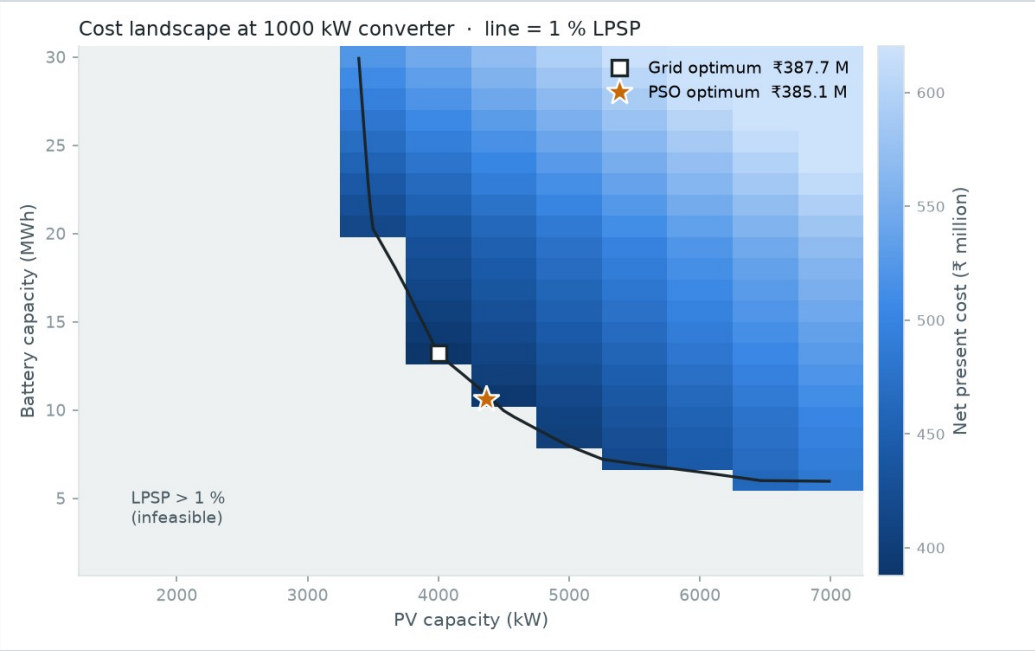


Where the NPC goes: capital (blue), 15-year costs (orange), salvage credit (green).

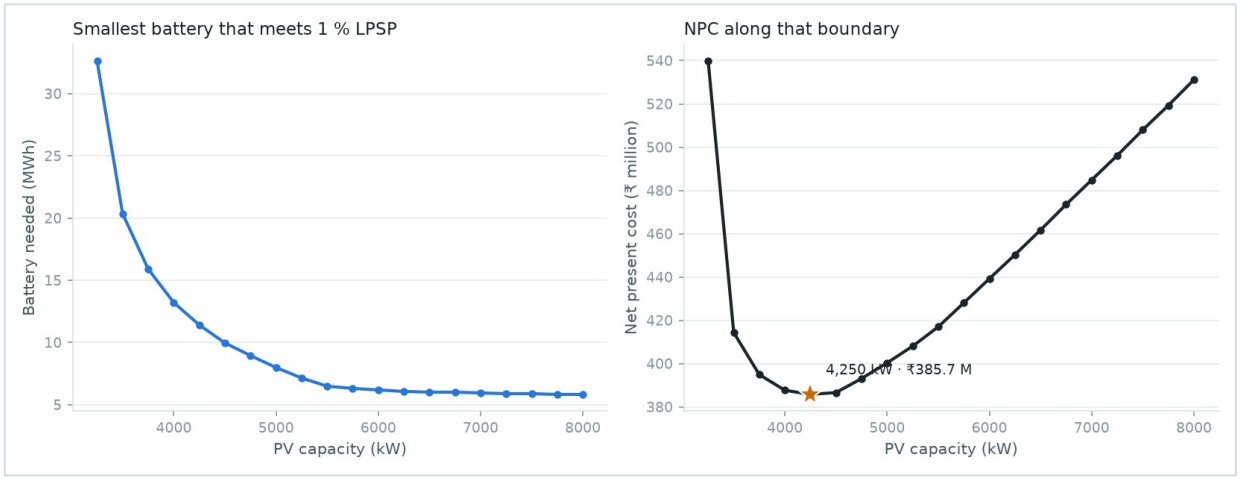
Takeaway **PSO ₹385.1 M vs grid sweep ₹387.7 M:** the same valley found two independent ways — strong evidence, not a proof of global optimality.

RESULTS

Design space: a flat valley along the reliability boundary



NPC landscape at 1,000 kW converter; line = 1 % LPSP.



Battery needed and NPC along the 1 % boundary.

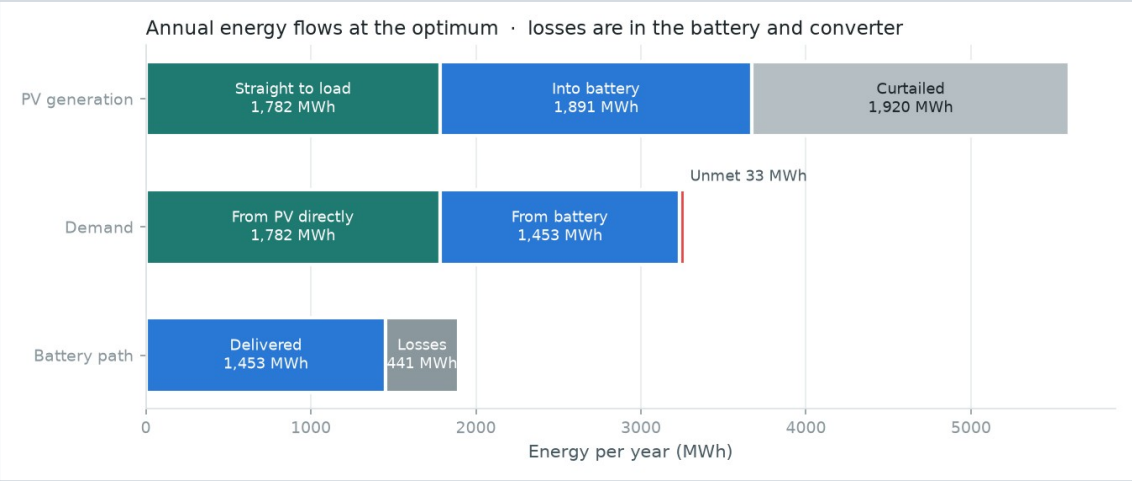
6
feasible grid designs within 2 % of the cheapest

1.6 %
cost spread of the five cheapest

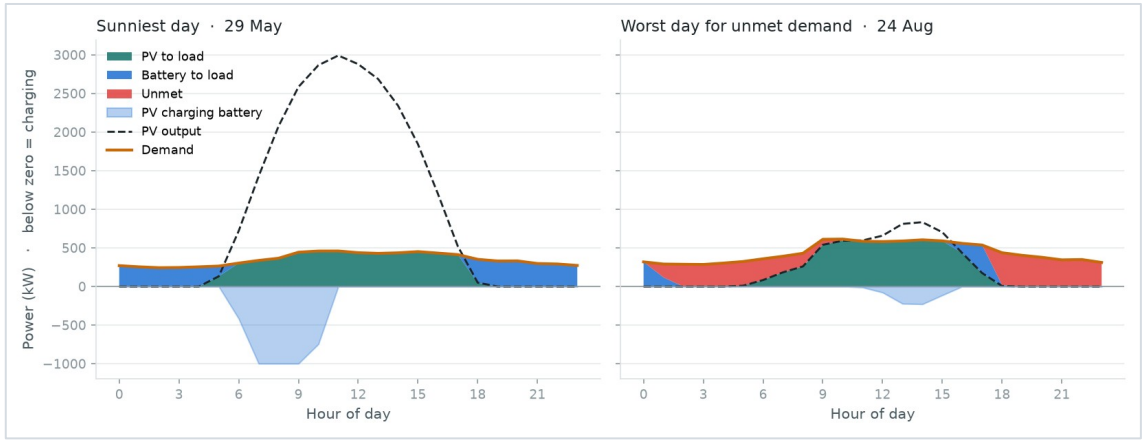
On the boundary
every optimum sits exactly at LPSP = 1 %

RESULTS

Energy flows and hourly dispatch



Annual energy flows at the optimum (MWh).



Hourly dispatch: sunniest day and worst day.

5,593 MWh
PV generated per year

34 %
of PV curtailed (battery full)

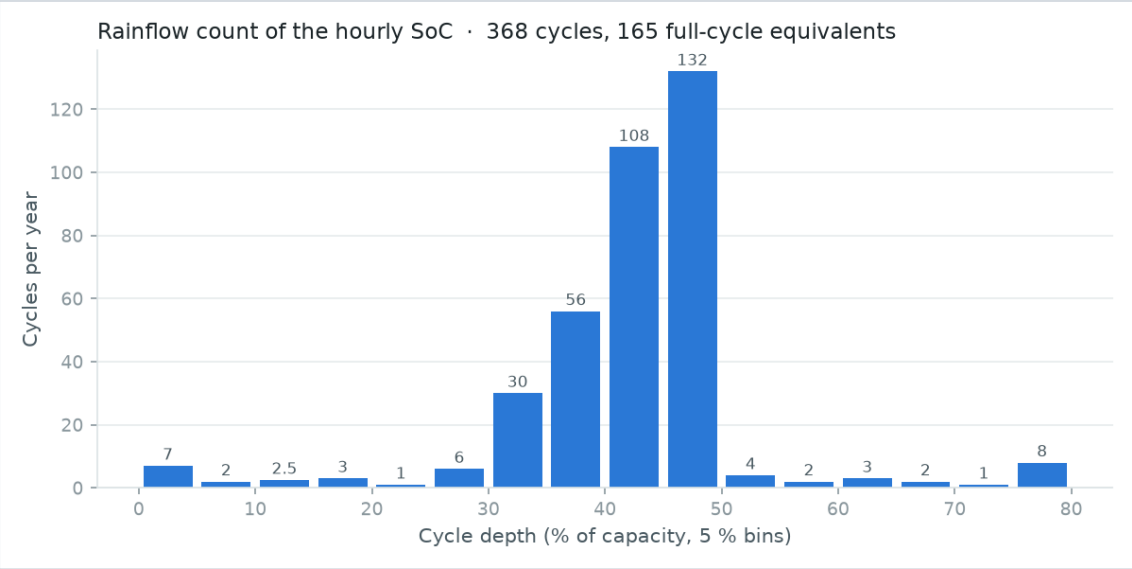
441 MWh
lost in battery and converter

33 MWh
unmet = 1 % of demand

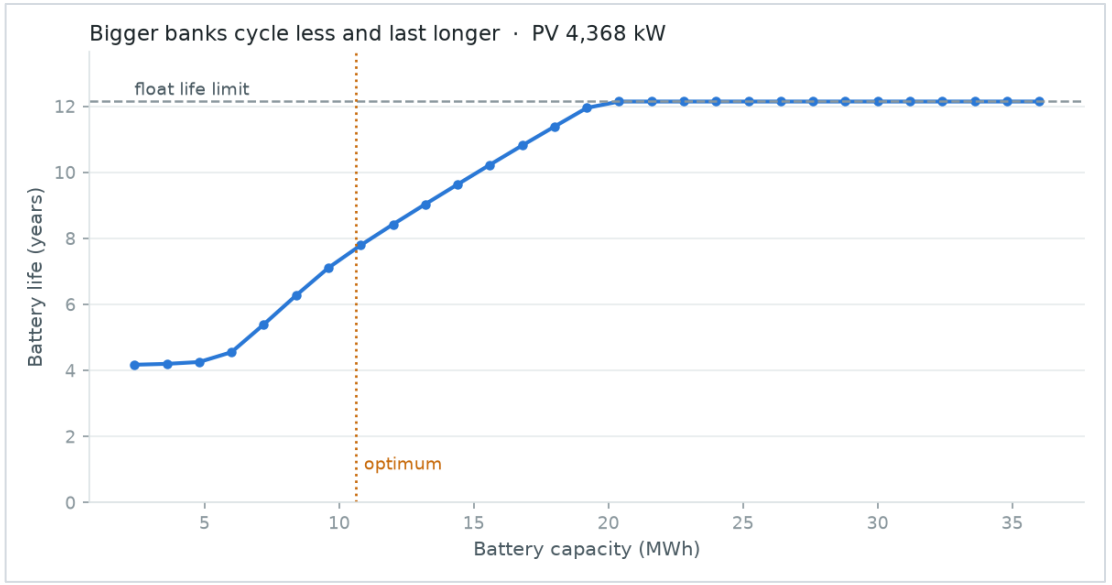
Takeaway An islanded system must oversize PV and throw energy away to stay reliable through dull spells.

RESULTS

Battery operation and ageing



Rainflow cycle depths of the hourly SoC (one year).



Battery life against bank size at the optimum PV.

7.7 y
wear life: cycling-limited

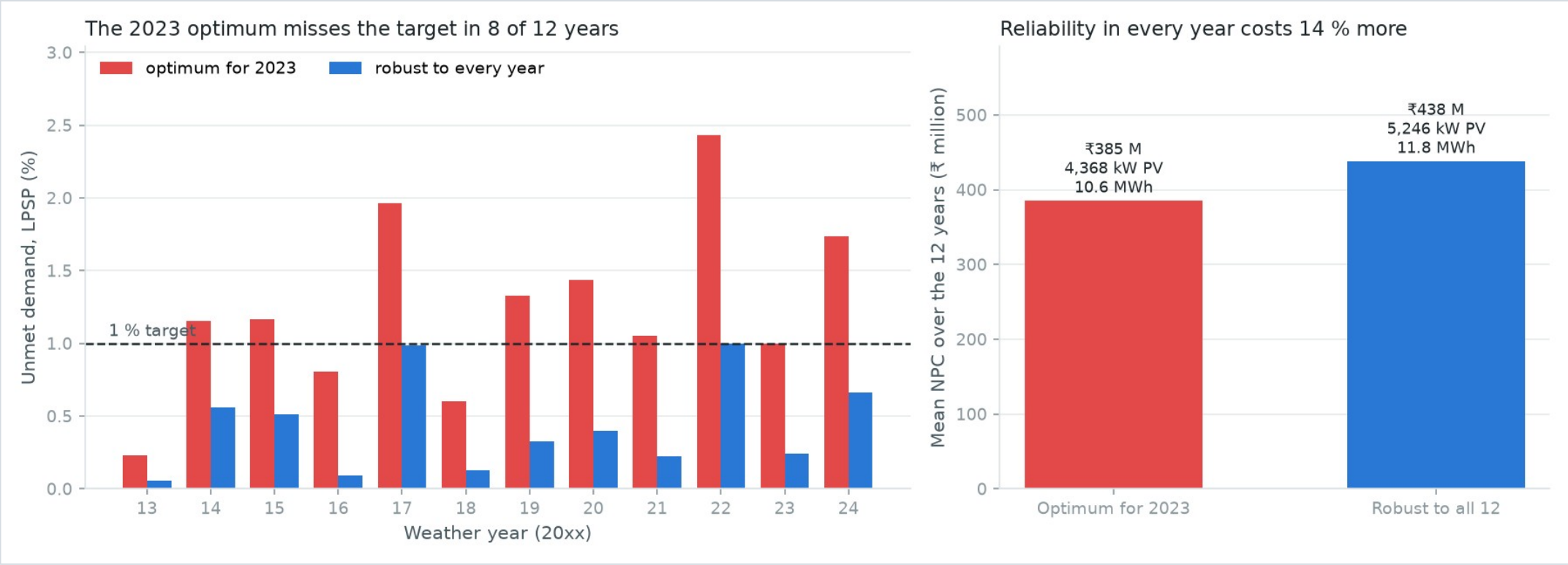
12.1 y
float life at site temperatures

year 7.7
one replacement in 15 years

₹54.6 M
replacements (present value)

Takeaway Ageing links sizing to cost: a larger bank cycles less, lasts longer and needs fewer replacements.

Multi-year weather robustness



Left: LPSP of each design in each weather year. Right: mean NPC over the 12 years.

8 of 12
years the 2023 design misses 1 %

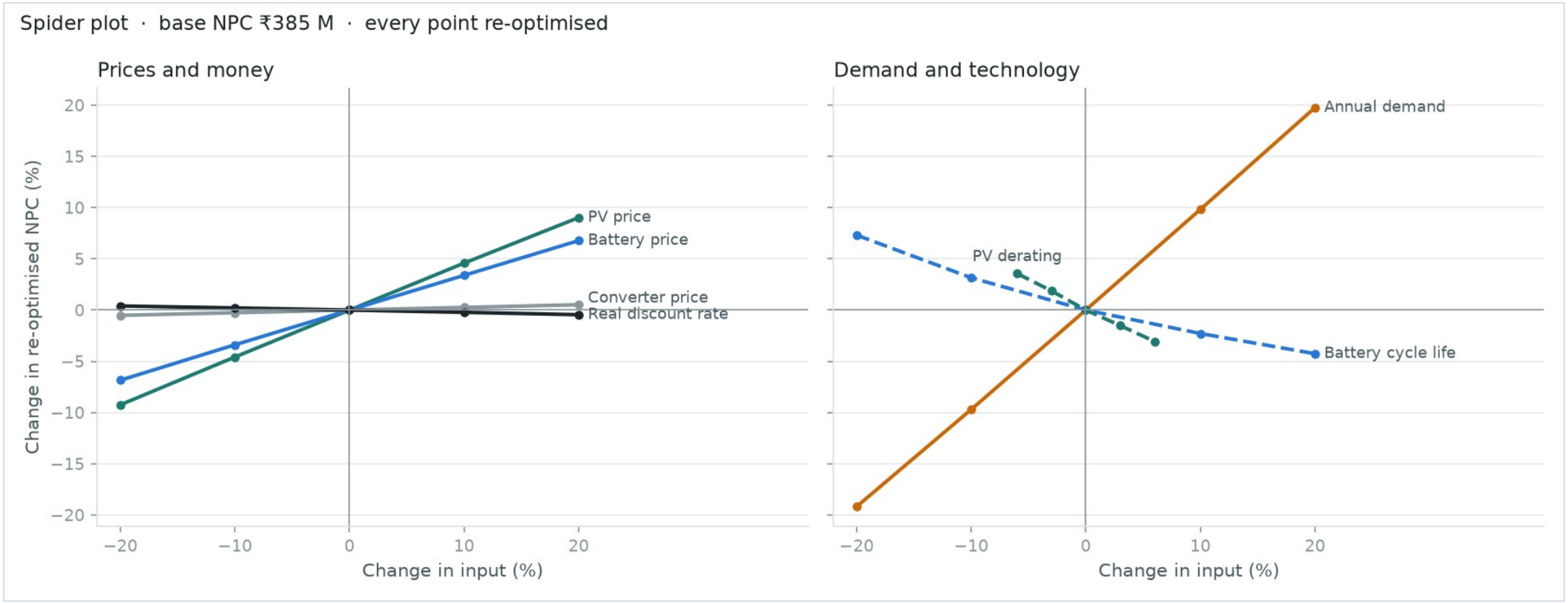
2.43 %
worst year (2022)

+14 %
cost of meeting 1 % in all 12 years

133 h
shortage hours in 2023 (17 events)

SENSITIVITY

One-at-a-time sensitivity of the re-optimised cost



Spider plot: % change of re-optimised NPC (base ₹385 M) for each input changed alone.

-19 / +20 %
demand −20 / +20 %

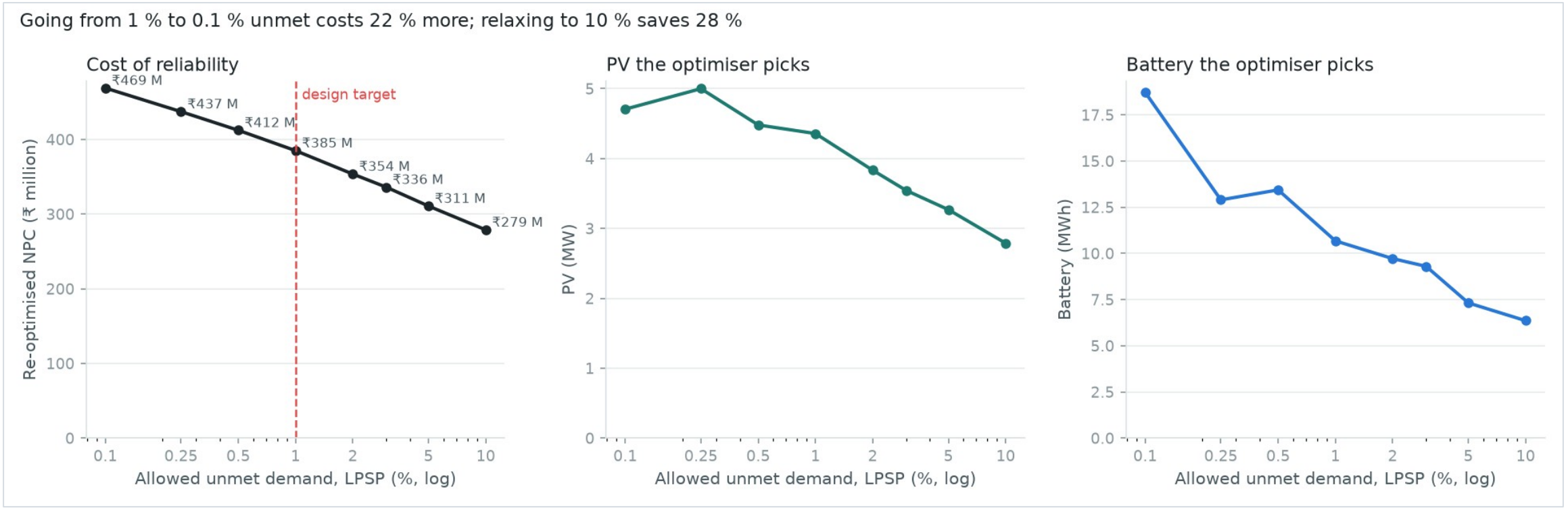
-9 / +9 %
PV price −20 / +20 %

-7 / +7 %
battery price −20 / +20 %

+7 / -4 %
cycle life −20 / +20 %

SENSITIVITY

The reliability target: what each step costs



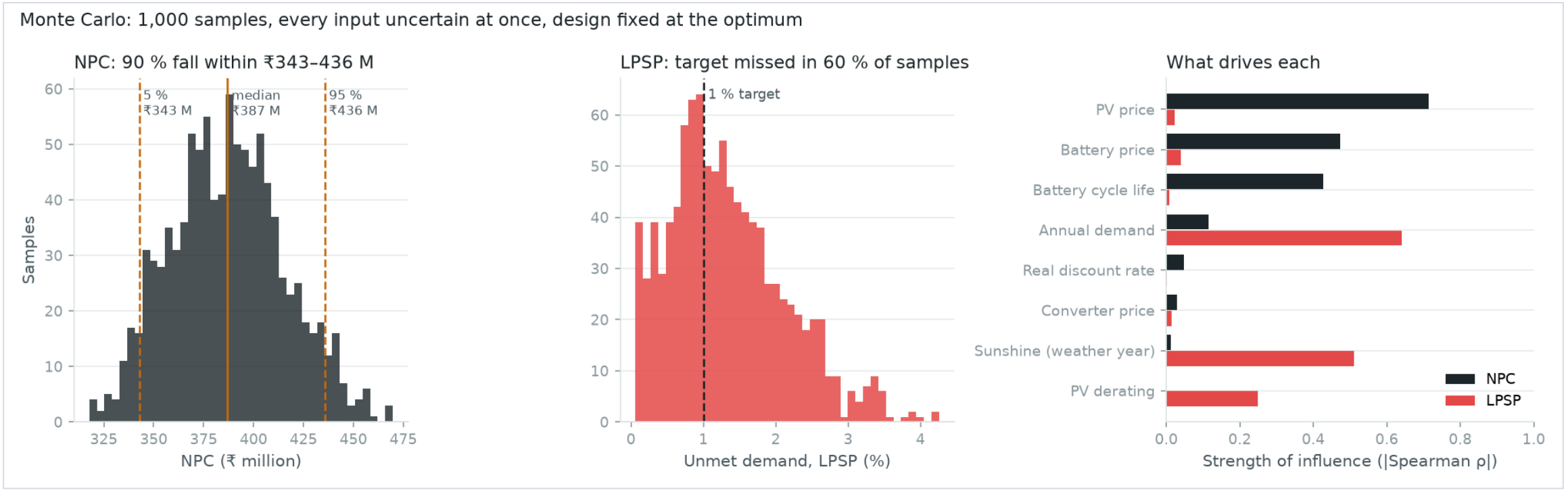
Re-optimised NPC, PV and battery against the allowed unmet demand (log scale).

+22 %
1 % → 0.1 % unmet (₹469 M)

₹385.1 M
design target: 1 % unmet

-28 %
1 % → 10 % unmet (₹279 M)

Monte Carlo uncertainty analysis



Distributions of NPC and LPSP, and the influence of each input (|Spearman ρ|).

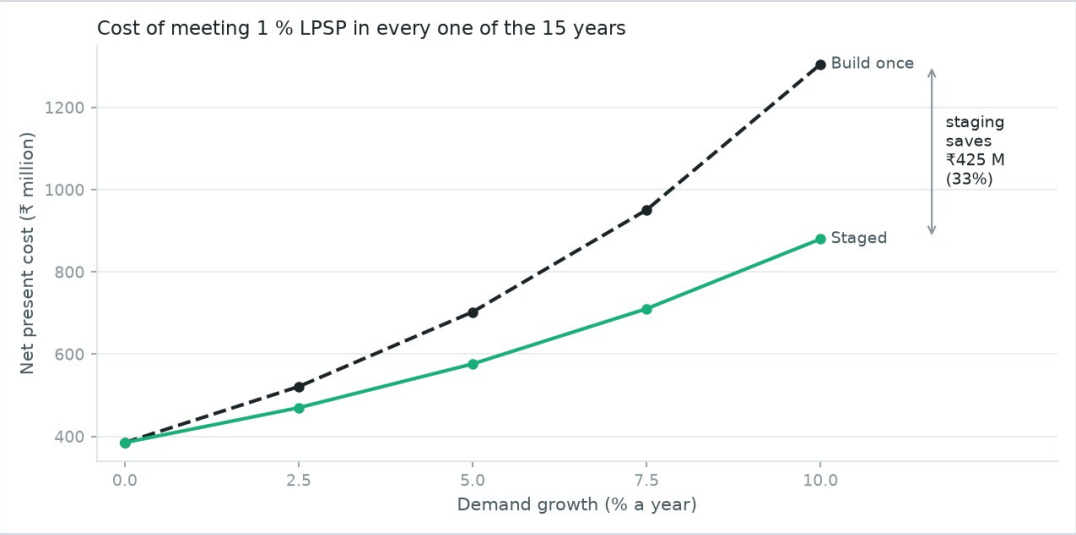
Sampled input	Distribution
PV, battery, converter prices	uniform ±20 %
Real discount rate, cycle life	uniform ±20 %
PV derating · demand	±6 % · ±10 %
Weather year	uniform, 2013–2024

₹343 M–436 M
5th–95th percentile of NPC

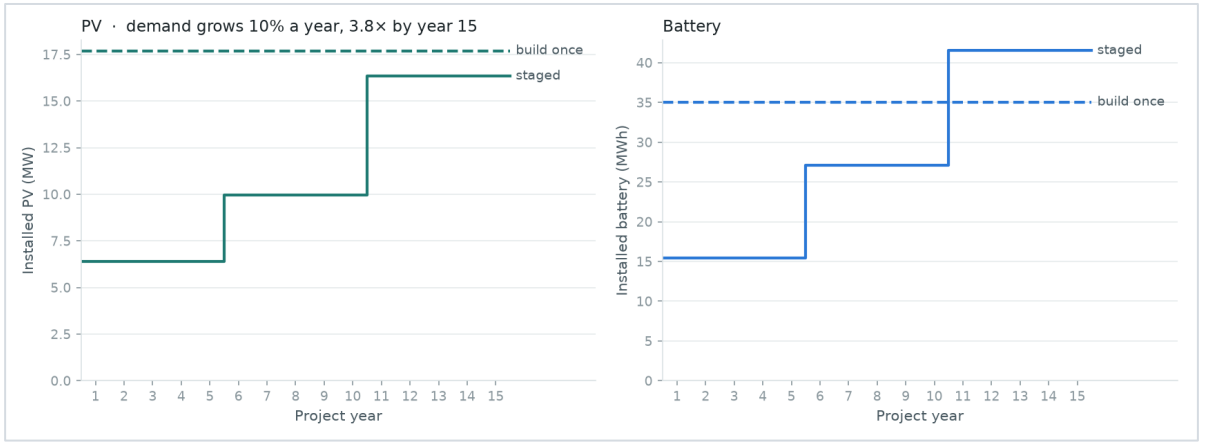
60 %
of samples miss the 1 % target

SCENARIOS

Load growth and staged expansion



NPC against growth rate.



Installed PV and battery, year by year (10 %/yr).

₹1,305 M
build once for year 15

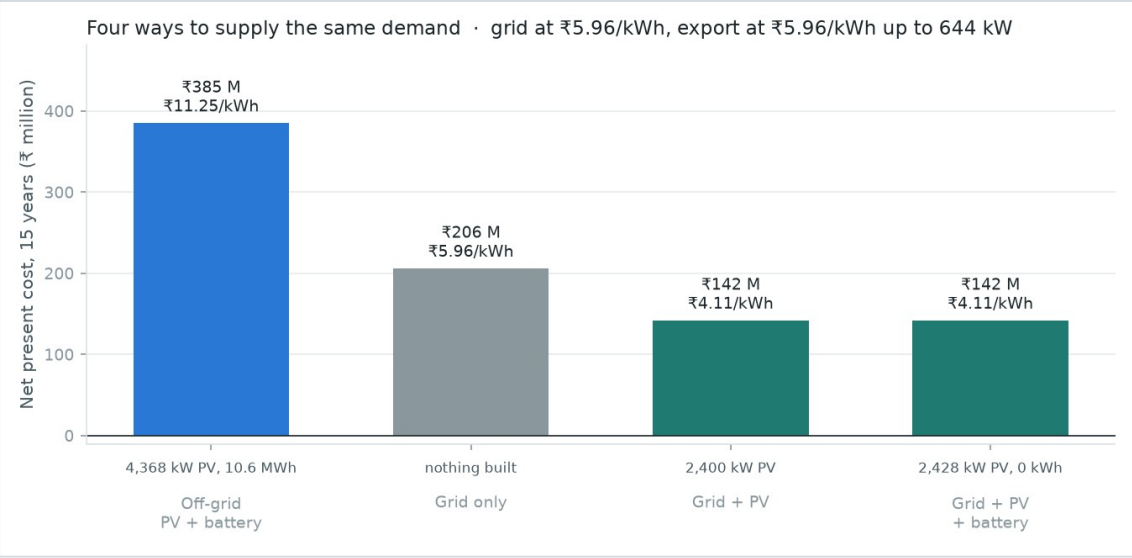
₹880 M
staged: years 1, 6, 11

−33 %
lifecycle cost saved

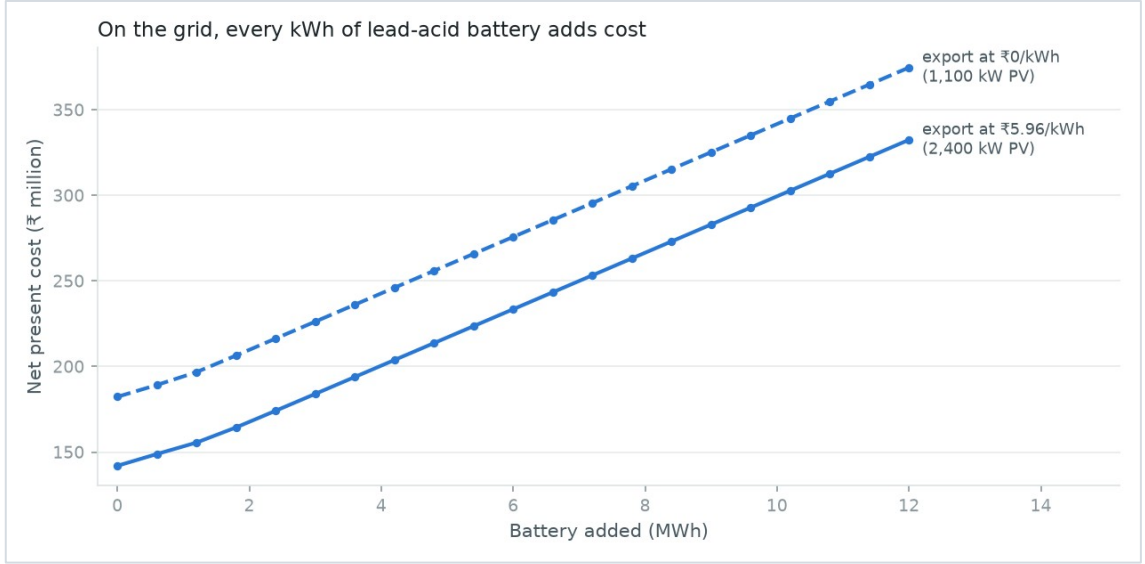
₹13.08/kWh
staged LCOE (paper: 11.96–18.47)

Takeaway Under growth, build in stages: later spending is discounted and capacity is not idle for years.

Grid-connected alternatives



Four ways to supply the same demand.



NPC as battery is added on the grid.

Assumptions

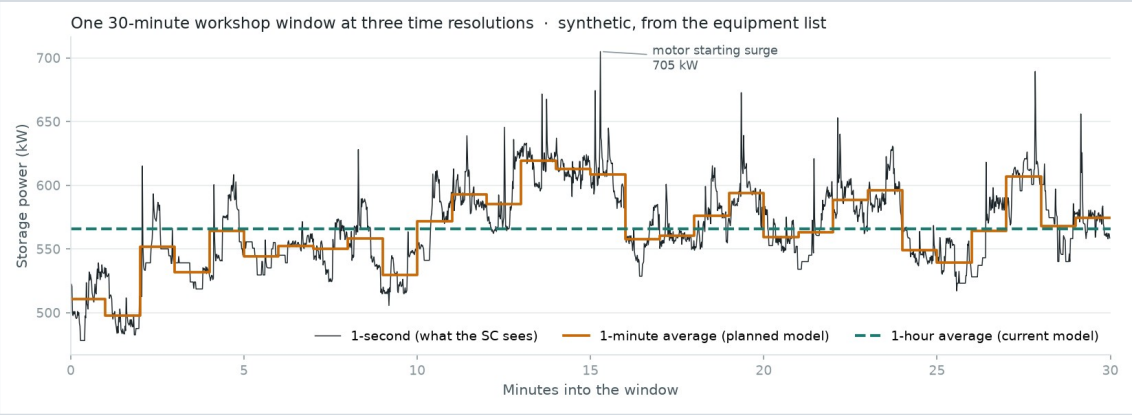
- Buy ₹5.96/kWh; export ₹5.96/kWh up to 644 kW
- Grid always available: no outages modelled

₹142 M
grid + 2,400 kW PV (₹4.11/kWh)

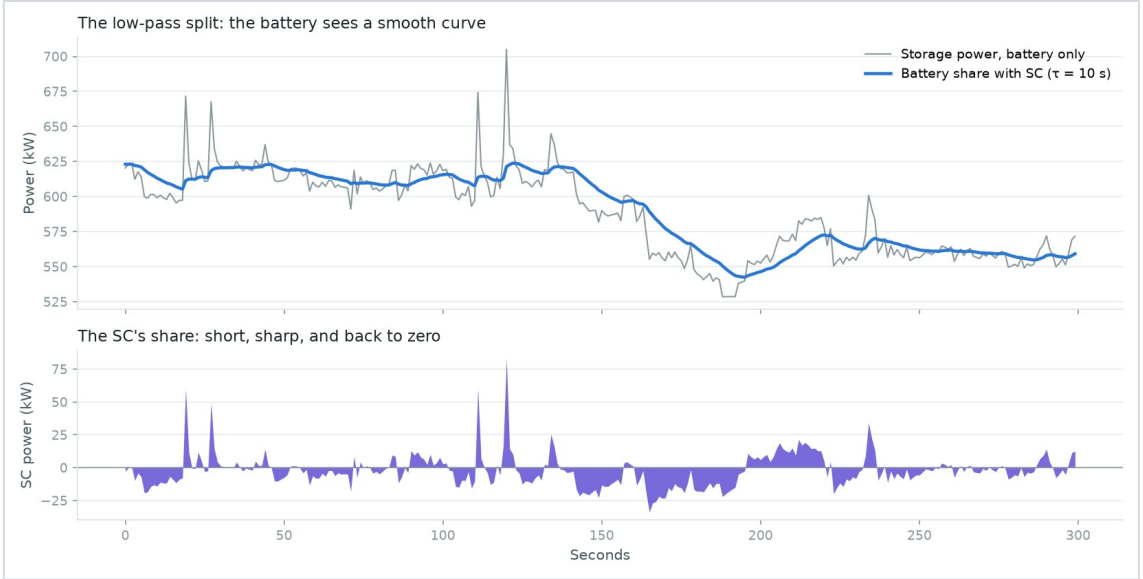
0 kWh
battery chosen at every export price

Takeaway With a reliable grid, storage is not worth it: the grid is the cheapest battery.

Supercapacitor: what it would see



Synthetic 1-s workshop load (from the equipment list).



Low-pass split: battery share (blue) and SC share (purple), τ = 10 s.

93 kW
coincident step, 5 s (465 kJ)

0.0088 C
step relative to the 10.6 MWh bank

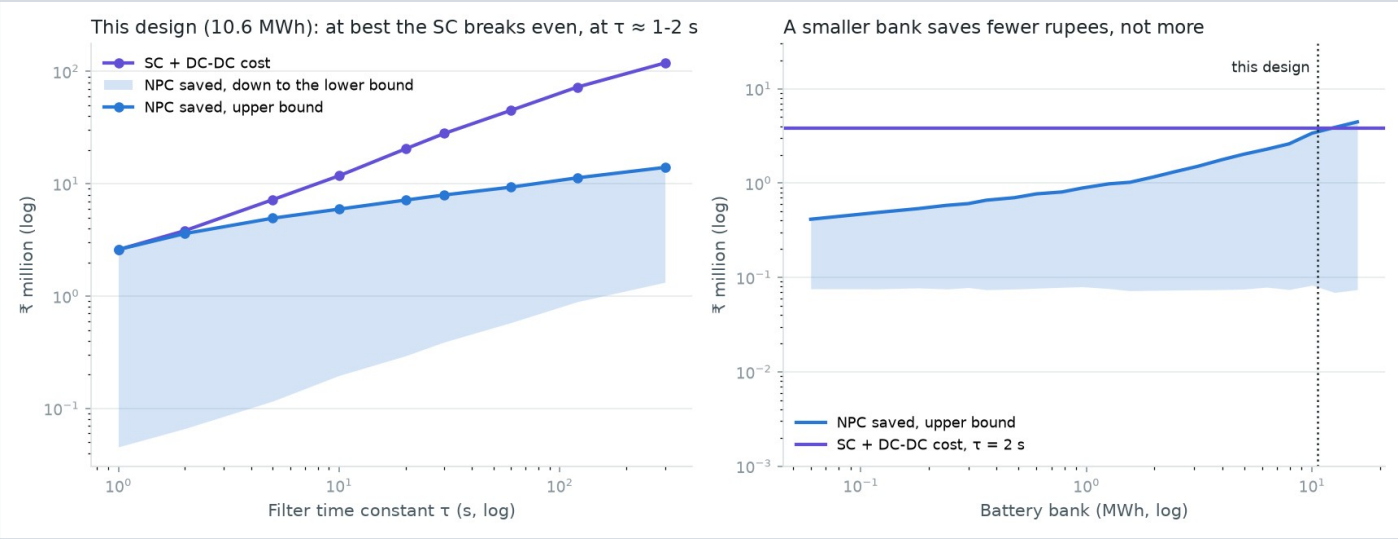
34 %
battery ramp left with τ = 2 s

4 × 24
strings × 16 V / 58 F modules

Takeaway Technically the SC smooths the battery's fastest ramps; the question is whether that pays.

SUPERCAPACITOR

Is a supercapacitor economically justified?



SC + DC–DC cost against battery NPC saved, by filter τ and by bank size.

Filter τ	SC + DC–DC cost	NPC saved (lower–upper)
1 s	₹2.59 M	₹0.05–2.60 M
2 s	₹3.82 M	₹0.07–3.60 M
10 s	₹11.77 M	₹0.20–5.94 M

Break-even at best
only with the upper bound, τ ≈ 1–2 s

0.0088 C
the step is no stress for this bank

Unresolved assumptions

- Module price ₹45,000 (placeholder)
- Motor starting surge 5× rated
- Machine on-times and power ratings

Takeaway Not justified on cost for this design. Revisit with real prices, or for small banks whose current limit the surge exceeds.

SOURCE VALIDATION

Checking the reference paper against CERC

Input	Kumar et al. (2018)	CERC order (primary)	Used in this project
PV capital cost	₹50,502 / kW	₹53,002 / kW (₹530.02 lakh/MW)	Paper value (base)
Discount factor	10.70 %	10.64 %	Paper value (base)

+2.3 %
NPC with CERC values

₹393.9 M
vs base ₹385.1 M

Unchanged
re-optimised design: same sizes

Sources

CERC, Petition SM/03/2016, order 29 Apr 2016, para 9 (discount factor) and para 27 (PV cost); CERC, Petition 17/SM/2015, order 23 Mar 2016 (PV cost breakdown). Both PDFs are in resources/.

DISCUSSION

Integrated engineering discussion

Cost

Demand ±20 % → ±20 % NPC; then PV price, battery price, cycle life.

Reliability

Nominal optimum sits on the 1 % edge; misses it in 8 of 12 years.

Robustness

+14 % buys 1 % in every weather year; Monte Carlo misses 60 %.

Demand growth

Staged expansion saves 33 % at 10 %/yr.

Grid access

Grid + PV is 63 % cheaper than off-grid; no battery.

Supercapacitor

Technical benefit is real; economic benefit is at best break-even.

Optimal microgrid sizing is a trade-off between **lifecycle cost, reliability and robustness** to changing operating conditions.

Conclusions, limitations and future work

Conclusions

- Optimum: 4,368 kW PV, 10.6 MWh, 998 kW; ₹385.1 M, ₹11.25/kWh
- PSO matches and improves on grid search (5/5 seeds)
- Robust to 12 weather years: +14 % cost
- Staging saves 33 % under 10 % growth
- Grid + PV: ₹142 M, no battery
- Supercapacitor: at best break-even

Limitations

- Synthetic load, not metered
- Verified (35 checks), not externally validated
- SC price, surge, on-times are placeholders
- Costs from 2016–17; lead-acid only
- Grid case ignores outages
- Hourly model; one load shape for all years

Future work

- Calibrate with department meter data
- Adopt the multi-year design as the main result
- Compare LFP batteries at today's prices
- Add diesel backup, as in Kumar et al.
- CO_2 savings and cost–emission trade-off
- Supplier quotes for SC and DC–DC stage

References and acknowledgements

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